

Global Health & Life Expectancy

An Analysis of World Development Indicators (1960-2024)

AnalystLab Africa — Data Analytics Internship (Batch B)

Capstone Project | Final Report

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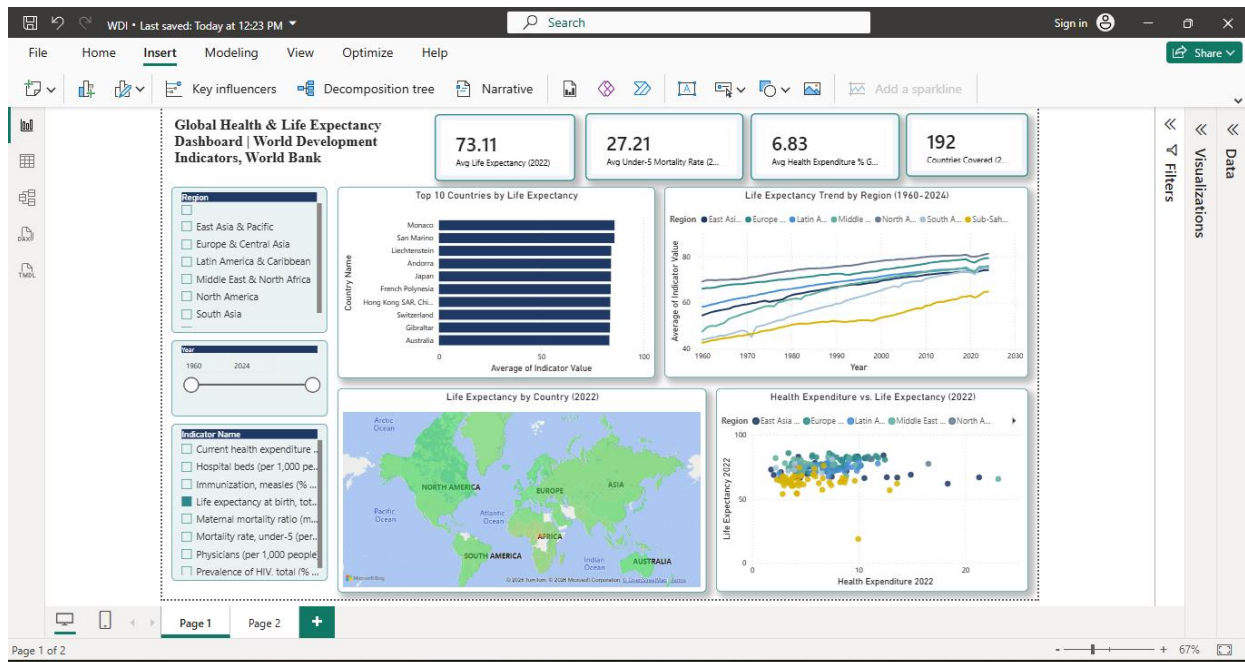


Figure 0: Final interactive Power BI dashboard

1. Project Objective / Problem Statement

This project applies the full data analytics workflow to the World Bank's World Development Indicators (WDI) dataset, with a focus on global health and life expectancy outcomes. The objective is to identify patterns in health outcomes and healthcare investment across countries and regions from 1960 to 2024, and to translate those patterns into insights and recommendations that could support health policy and resource allocation decisions.

The specific problem addressed is: **how do life expectancy and other key health outcomes vary across countries and regions, and to what extent does healthcare spending explain that variation?**

2. Dataset Description

The dataset used is the World Bank's World Development Indicators (WDI), a global collection of development statistics covering over 200 countries and territories. The full bulk download (WDI_CSV.zip) contains 400,000+ rows across all available indicators, countries, and years (1960-2025) in a wide format (one column per year).

From this, 8 indicators relevant to the Health & Life Expectancy theme were selected:

Indicator	Code
Life expectancy at birth, total (years)	SP.DYN.LE00.IN
Mortality rate, under-5 (per 1,000 live births)	SH.DYN.MORT
Maternal mortality ratio (per 100,000 live births)	SH.STA.MMRT
Current health expenditure (% of GDP)	SH.XPD.CHEX.GD.ZS
Physicians (per 1,000 people)	SH.MED.PHYS.ZS
Hospital beds (per 1,000 people)	SH.MED.BEDS.ZS
Prevalence of HIV, total (% of population ages 15-49)	SH.DYN.AIDS.ZS
Immunization, measles (% of children ages 12-23 months)	SH.IMM.MEAS

Country-level metadata (region, income group) was sourced from the accompanying WDIcountry.csv file and merged into the main dataset to distinguish real countries from regional and income-group aggregates (e.g., "World", "Sub-Saharan Africa", "High income").

3. Data Cleaning Process

1. Loaded the raw WDICSV.csv (397,000+ rows) into Power Query and filtered to the 8 selected indicator codes using a Table.SelectRows/List.Contains formula (the standard checkbox filter could not surface all codes due to Power Query's 1,000-value dropdown sampling limit).
2. Unpivoted the year columns (1960-2025) from wide format into a long format with Country, Indicator, Year, and Value columns, enabling proper time-series analysis.
3. Removed rows with null indicator values to retain only genuine data points, resulting in a clean set of 73,246 rows.

4. Merged in country metadata (Region, Income Group) from WDIcountry.csv via a left-outer join on Country Code.
5. Created an "Entity Type" flag (Country vs. Aggregate) based on whether a Region value was present, allowing regional/income-group aggregates (e.g., "World", "Arab World") to be excluded from country-level analysis while remaining available for reference.

Data Quality Finding

During analysis, an anomalous value was identified for the Central African Republic's 2022 life expectancy, recorded in the WDI dataset as 18.8 years. This figure is inconsistent with the country's historical trend (low-to-mid 50s throughout the preceding decade) and with external sources reporting approximately 54.5 years for the same year. Rather than exclude or silently correct this value, it was retained in the dashboard and is flagged here for transparency. It illustrates the importance of cross-referencing outlier values against historical trends and external sources before drawing conclusions, even authoritative global datasets can contain reporting errors.

4. Analysis Methodology

2022 was selected as the primary reference year for point-in-time comparisons (KPIs, bar chart, map, scatter plot), rather than the most recent year (2024), because 2024 data coverage was substantially incomplete (only ~950 indicator/country/year data points recorded, versus ~1,700 for 2022), using 2024 would have produced misleading averages reflecting only the subset of countries that had reported early. The full 1960-2024 range was retained for trend analysis, where completeness matters less than long-run pattern.

All country-level aggregates (Entity Type = Country) were isolated from regional/income aggregates throughout the analysis to avoid double-counting or skewed averages. Two DAX measures (Life Expectancy 2022 and Health Expenditure 2022) were created to enable a paired country-level comparison for the scatter plot.

5. Key Findings

Dashboard Overview

The dashboard (Figure 0) presents four summary KPIs anchored to 2022 (Average Life Expectancy: 73.11 years; Average Under-5 Mortality Rate: 27.21 per 1,000; Average Health Expenditure: 6.83% of GDP; Countries Covered: 192), alongside interactive charts: a regional life expectancy trend (1960-2024), a Top 10 countries bar chart, a world map shaded by life expectancy, and a health-expenditure-vs-life-expectancy scatter plot, all filterable by Region, Year, and Indicator via slicers.

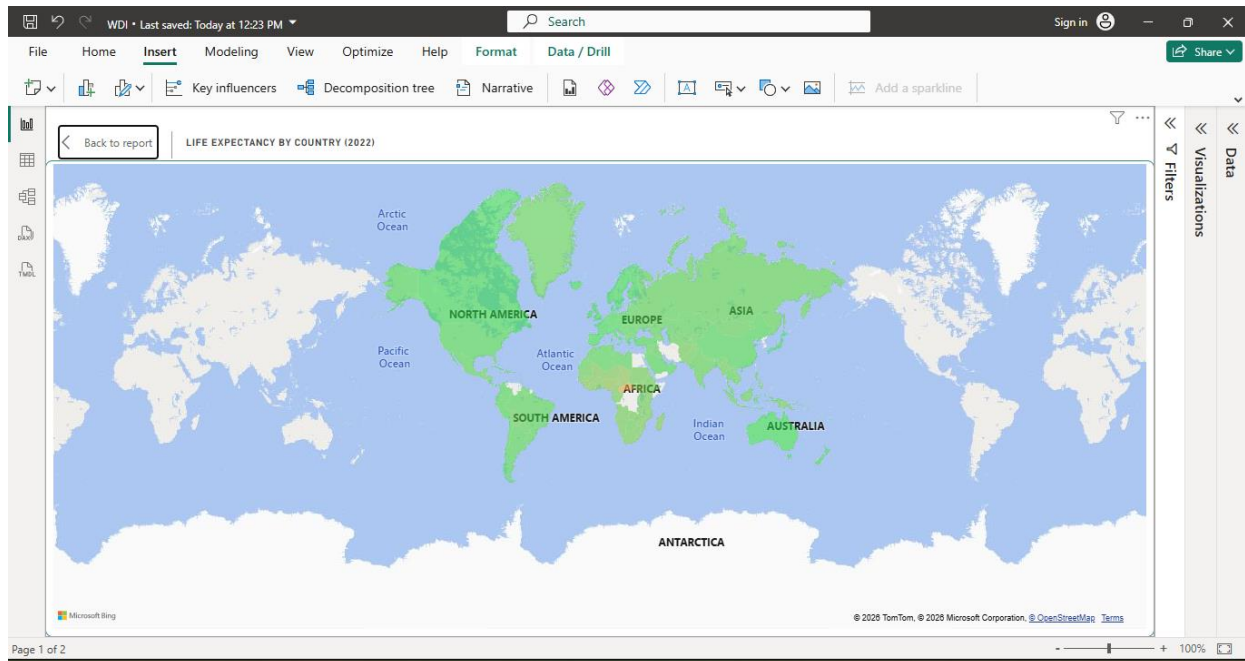


Figure 1: Life expectancy by country, 2022 (darker green = higher life expectancy)

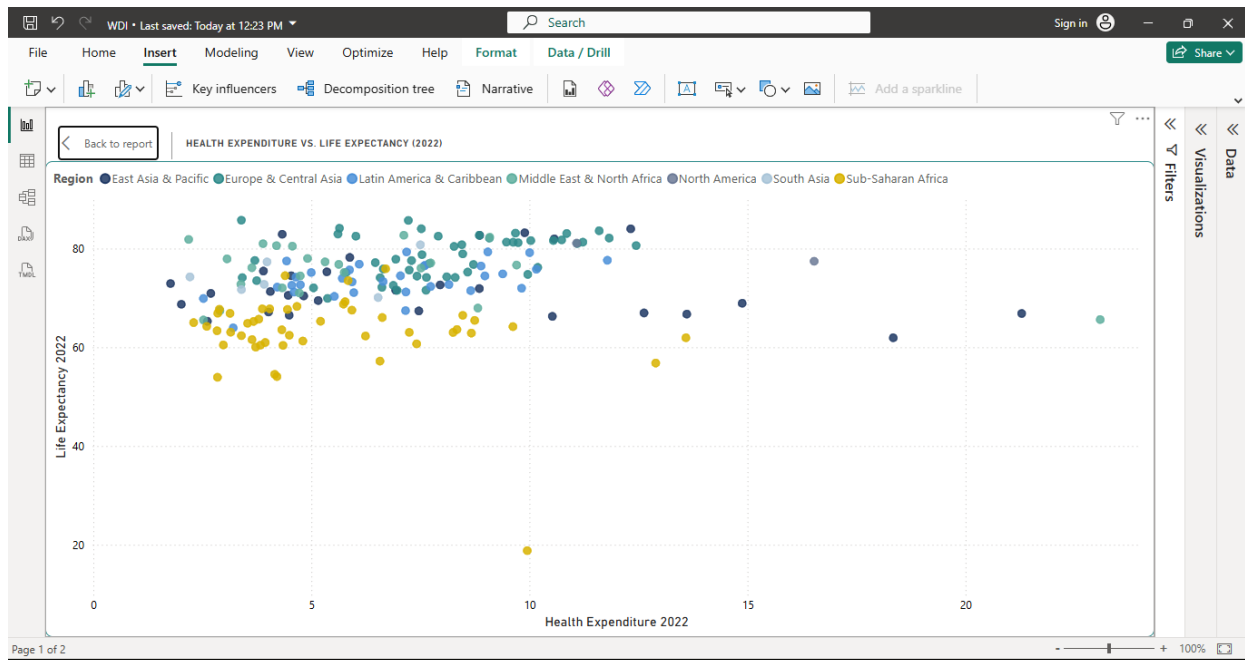


Figure 2: Health expenditure (% GDP) vs. life expectancy, 2022, colored by region

- Global average life expectancy stood at 73.11 years in 2022, but this masks sharp regional disparities. Sub-Saharan Africa consistently trails other regions by 15-20+ years across the 1960-2024 trend (Figure 0, trend chart).
- Health expenditure and life expectancy show a positive but plateauing relationship: gains are steep up to roughly 6-10% of GDP spent on health, after which additional spending yields diminishing returns (Figure 2).
- The 10 countries with the highest life expectancy (Monaco, San Marino, Liechtenstein, Andorra, Japan, French Polynesia, Hong Kong SAR, Switzerland, Gibraltar, Australia) are predominantly small, wealthy, or highly developed economies, suggesting factors beyond raw health spending (income level, healthcare system quality, lifestyle) matter significantly.
- Data quality matters as much as data quantity: the Central African Republic anomaly (Section 3) demonstrates that even authoritative global datasets can contain reporting errors, reinforcing the need for validation before drawing conclusions.

6. Recommendations

1. For policymakers in lower-life-expectancy regions: prioritize healthcare system efficiency and access rather than raw spending increases alone, since the data shows spending beyond ~10% of GDP does not proportionally translate into longer life expectancy.
2. For global health organizations: target interventions at Sub-Saharan Africa specifically, where the gap to global averages remains largest and has persisted across six decades of data.
3. For data users and analysts: always cross-validate outlier values in global datasets against historical trends and secondary sources before drawing conclusions, as demonstrated by the Central African Republic data anomaly identified in this analysis.

7. Conclusion

This project walked through the complete data analytics workflow from raw, unwieldy source data through cleaning, transformation, and interactive dashboard design, to actionable insight. The resulting Health & Life Expectancy dashboard shows that while global health outcomes have improved substantially since 1960, meaningful regional disparities remain, and healthcare spending alone is an incomplete predictor of outcomes. Equally important, the project surfaced a genuine data quality issue within the source dataset itself, underscoring that careful validation is a core part of any credible analysis, not an optional step.

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